

QUESTION 1

Financial File Compression System Using Divide and Conquer

Aim

To simulate a financial file compression system using the Divide and Conquer technique and analyze its time complexity.

Problem Statement

A financial organization stores large files containing transaction records. To improve processing speed, the file is divided into smaller blocks and compressed recursively. The objective is to implement this approach and analyze its efficiency.

Algorithm

1. Read the number of file blocks.
2. Divide the file into two smaller parts recursively.
3. Continue dividing until a single block remains.
4. Compress each block individually.
5. Display the compression process.
6. Analyze the time and space complexity.

Source Code

```
#include <stdio.h>
```

```
void compressFile(int start, int end)
```

```
{
```

```
    if(start == end)
```

```
    {
```

```
        printf("Compressing block %d\n", start);
```

```
        return;
```

```
    }
```

```
int mid = (start + end) / 2;

compressFile(start, mid);
compressFile(mid + 1, end);
}

int main()
{
    int blocks;

    printf("Enter number of file blocks: ");
    scanf("%d", &blocks);

    compressFile(1, blocks);

    printf("\nTime Complexity : O(n)\n");
    printf("Space Complexity: O(log n)\n");

    return 0;
}
```

```

1  #include <stdio.h>
2
3  void compressFile(int start, int end)
4  {
5      if (start == end)
6      {
7          printf("Compressing block %d\n", start);
8          return;
9      }
10
11     int mid = (start + end) / 2;
12
13     compressFile(start, mid);
14     compressFile(mid + 1, end);
15 }
16
17 int main()
18 {
19     int blocks;
20
21     printf("Financial File Compression System\n");
22     printf("-----\n");
23
24     printf("Enter number of file blocks: ");
25     scanf("%d", &blocks);
26
27     compressFile(1, blocks);
28
29     printf("\nAnalysis:\n");
30     printf("Recurrence Relation :  $T(n) = 2T(n/2) + O(1)$ \n");
31     printf("Time Complexity      :  $O(n)$ \n");
32     printf("Space Complexity       :  $O(\log n)$ \n");
33
34     return 0;
35 }

```

Input

Number of file blocks = 8

Output

Compressing block 1

Compressing block 2

Compressing block 3

Compressing block 4

Compressing block 5

Compressing block 6

Compressing block 7

Compressing block 8

Time Complexity : $O(n)$

Space Complexity : $O(\log n)$

Financial File Compression System

Enter number of file blocks: 8

Compressing block 1

Compressing block 2

Compressing block 3

Compressing block 4

Compressing block 5

Compressing block 6

Compressing block 7

Compressing block 8

Analysis:

Recurrence Relation : $T(n) = 2T(n/2) + O(1)$

Time Complexity : $O(n)$

Space Complexity : $O(\log n)$

=== Code Execution Successful ===

Complexity Analysis

- Time Complexity = $O(n)$
- Space Complexity = $O(\log n)$
- Recursion follows the Divide and Conquer paradigm.

Result

Thus, the financial file compression system was implemented successfully using Divide and Conquer. The algorithm efficiently processes large files with linear time complexity.